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Public Acceptance of an Increase in the Mandatory Biodiesel Blending Ratio: An Evaluation to Guide Policymaking in South Korea

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Abstract

South Korea has decided to increase the mandatory biodiesel (BD) blending ratio from 5.0% to 8.0% in 2030 to mitigate CO_2 emissions. This study provides a nationally representative empirical estimate of public willingness to pay (WTP) specifically for increasing the mandatory BD blending ratio, and addresses a critical gap in the literature on biofuel policy acceptance. Although a one-and-one-half-bound format was employed, the single-bound spike model is adopted as the main specification due to evidence of response effects. This paper seeks to delve into public acceptance of the increase by gathering and analyzing the data on the public's WTP through the application of contingent valuation (CV). Based on a national CV survey administered over a five-week period from mid-April to mid-May 2025, 1000 valid observations were obtained and analyzed. Since 60.5% of all respondents stated a WTP of 0, a spike model that could account for this was adopted. The key coefficients and the model achieved statistical significance. The average household WTP figure obtained was KRW 5052.3 (USD 3.50) per annum. Expanding this value to the entire population gives us KRW 111.69 billion (USD 77.46 million) annually, based on December 2024 constant prices. The additional costs in 2030 resulting from the increase will reach KRW 456.24 billion (USD 316.41 million). Since the WTP figure is smaller than these additional costs, it seems that public acceptance is not sufficiently high. Therefore, it is necessary to implement policy measures such as government subsidies and research and development support to reduce the price of BD.

Keywords: biodiesel; blending ratio; public acceptance; willingness to pay; contingent valuation

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1. Introduction

The transportation sector accounts for approximately 17% of global greenhouse gas (GHG) emissions and 25% of carbon dioxide (CO_2) emissions, as per the International Energy Agency [1], with diesel vehicles emitting 7% more CO_2 per mile than gasoline equivalents. While electric and hydrogen fuel cell vehicles offer zero tailpipe emissions, high costs and infrastructure gaps—exacerbated by recent electric vehicle market chasms—ensure internal combustion engines (ICEs) retain a significant market share through 2030 [2]. Greening ICE fuels via biofuel blending, thus, remains urgent for nearterm decarbonization, particularly in the case of diesel, where biodiesel (BD) reduces lifecycle CO_2 and pollutants [3–5].

BD is a renewable fuel produced through the transesterification process, wherein vegetable oils, animal fats, or recycled greases react with methanol in the presence of a catalyst (typically sodium or potassium hydroxide) to yield fatty acid methyl esters and glycerol as a co-product. In South Korea, where edible oil production is limited, waste cooking oil (WCO)—primarily collected from restaurants and food processing facilities—serves as the predominant feedstock, accounting for approximately 85–90% of domestic BD production. In 2024, South Korea produced roughly 1.2 million kiloliters of BD annually, sufficient to meet the mandatory 4% BD blending requirement while supporting limited industrial applications. This reliance on WCO recycling aligns with United Nations Sustainable Development Goals 7 (Affordable and Clean Energy), 11 (Sustainable Cities and Communities), and 12 (Responsible Consumption and Production) as it simultaneously addresses waste management, energy security, and decarbonization of the transport sector—where diesel vehicles contribute 25% of national CO_2 emissions.

South Korea has mandated petroleum refiners to produce diesel fuel blended with BD in a certain proportion and to supply this blended fuel to gas stations in order to reduce CO_2 emissions from diesel vehicles [6]. From early 2024, the mandatory blending ratio has been set at 4.0%. The government initially announced its intention to increase the current mandatory ratio to 5.0% by 2030, aiming for a 1.0%-point increase. However, due to the upward revision of the national GHG-reduction target in the transportation sector at the end of 2022, the original target of 5.0% has been increased to 8.0%, a 3.0%-point increase. In summary, the current mandatory blending ratio of 5.0% for BD is scheduled to be increased to 8.0% by 2030. Since diesel is also used for industrial purposes, BD can be an alternative to natural gas for industrial purposes [7].

The government mandates refiners to blend BD at rising ratios: 4.0% (2024), escalating to 8.0% by 2030 (a 3.0-point increase from the initial 5.0% target, revised post-2022 NDC tightening). This policy—projected to require 633.6 million additional liters of BD amid 21.12 billion liters diesel demand—imposes KRW 456.24 billion in annual costs (wholesale premiums passed on via retail prices/subsidies), risking a backlash unless we can demonstrate public acceptance. Success hinges on quantifying household willingness to pay (WTP) for mandate tightening under realistic financing (e.g., income taxes funding residual costs post-pass-through), informing benefit–cost ratios against CO_2 abatement (1642 kt/year) and co-benefits. This study addresses this policy decision problem via contingent valuation (CV): delivering nationally representative WTP estimates for the explicit 5.0%→8.0% increase, distinguishing protest from true zeros via spike modeling, and benchmarking against costs to guide subsidies/phasing—pivotal for equitable decarbonization where diesel underpins 25% of national CO_2 emissions.

This study attempts to use WTP as an indicator for examining public acceptance. WTP is an economic term that means the maximum amount an individual intends to pay to consume or use a particular item of goods. Public acceptance constitutes a multifaceted construct in the energy transition literature, encompassing cognitive (risk/benefit perceptions), affective (emotional responses), normative (moral approval), and behavioral (support/willingness-to-act) dimensions essential for successful policy implementation. While attitudinal surveys capture broad acceptance through Likert-scale measures of approval or trust in institutions, such approaches often fail to quantify intensity of support under realistic economic trade-offs, limiting their utility for cost–benefit analysis in resource allocation decisions. This study adopts WTP—elicited through CV—as the primary empirical proxy for public acceptance, representing the standard economic approach where monetary commitments under budget constraints reveal the marginal value individuals place on policy outcomes.

Table 1 summarizes information from studies analyzing the public's WTP for biofuels. Jeanty and Hitzhusen [8] used CV to derive a price premium (PP) for diesel blended with 20% BD instead of conventional diesel. Solomon and Johnson [9] explored the PP for bioethanol produced from cellulose, an organic compound, over conventional gasoline through CV. Yabe [10] applied CV to measure the WTP for appropriate subsidies supporting domestic bioethanol production instead of the production of conventional gasoline. Petrolia et al. [11] used CV to assess the PP for the replacement of gasoline by a mixture of bioethanol and gasoline, with varying blending ratios.

Table 1. Summary of some previous studies investigating the public's willingness to pay regarding biofuels.

Sources	Objects to Be Valued	Countries	Main Findings	Methods [a]
Jeanty and Hitzhusen [8]	Blending biodiesel by 20% instead of diesel	United States	USD 0.14 to 0.26 per gallon over the price for diesel	CV
Solomon and Johnson [9]	Cellulosic bioethanol instead of gasoline	United States	USD 556 over annual expenditure per person on gasoline	CV
Yabe [10]	Domestic bioethanol instead of gasoline	Japan	JPY 3 per liter over the price for gasoline	CV
Petrolia et al. [11]	Blending bioethanol by 10% or 85% (E10 or E85) instead of gasoline	United States	USD 0.12 and 0.15 for E10 and E85 per gallon over the price for gasoline	CV
Huh et al. [12]	Enforcing renewable fuel standard scheme	South Korea	KRW 4432.9 per household per month	CV
Lim et al. [13]	Blending bioethanol by 5% instead of gasoline	South Korea	KRW 290 (USD 0.27) per liter over the price for gasoline	CV
Sivashankar et al. [14]	Biodiesel produced from Jatropha instead of diesel	Sri Lanka	INR 109 (EUR 0.74) per liter over the price for diesel	CV
Li and McCluskey [15]	Lignocellulosic bioethanol instead of gasoline	United States	11% of the price for gasoline	CV
Mamadzhanov et al. [16]	Lignocellulosic bioethanol instead of gasoline	South Korea	4.3% of the price for gasoline	CV
Kim et al. [17]	Four effects of marine biodiesel technology development	South Korea	KRW 1082.7 (USD 0.9), KRW 918.1 (USD 0.8), and KRW 258.3 (USD 0.2) per household per month for a 1%-point reduction in greenhouse gas emissions, a 1%-point mitigation of air pollutant emissions, and the creation of 100 new jobs	CE
Kim et al. [18]	Renewable methane instead of natural gas	South Korea	KRW 191.46 (USD 0.17) per m^3 over the price for natural gas	CV

Note: [a] CV and CE denote, respectively, contingent valuation and choice experiment.

Huh et al. [12] and Lim et al. [13] adopted CV for delving into the South Korean public's acceptance of supporting and introducing mandatory blending of biofuels. Sivashankar et al. [14] applied CV to evaluating diesel vehicle users' PP for BD produced from Jatropha over conventional diesel in Sri Lanka. Li and McCluskey [15] used CV to estimate the the PP for lignocellulosic bioethanol over conventional gasoline in South Korea. Mamadzhanov et al. [16] studied a similar topic in the United States. Kim et al. [17] quantified the marginal WTP values for four effects of marine BD technology development through a choice experiment (CE). They include reducing GHG emissions, mitigating air pollutant emissions, creating jobs, and improving energy security. Kim et al. [18] estimated the PP for renewable methane compared to natural gas by adopting CV.

Three implications emerge from the literature review presented above. First, to the best of our knowledge, this represents the first nationally representative CV study quantifying household WTP specifically for a mandatory BD blending ratio increase (from 5.0% to 8.0% by 2030 in South Korea), using a spike model to robustly handle high protest zeros. The previous studies have considered the WTP for the use of biofuels themselves, such as BD, bioethanol, and renewable methane, or for blending biofuels, but have not addressed the WTP for increasing the blending ratio. Second, as far as the authors are aware, this study is the first to explore public acceptance of increasing the BD blending ratio in South Korea. Third, the method used in this paper is consistent with that in former studies. Among a total of eleven cases presented in Table 1, ten utilized CV.

Despite these advances, a critical operational gap persists: the previous literature scarcely delivers nationally representative household WTP estimates for a specific mandate tightening—here, a 3.0 percentage-point BD increase from 5.0% to 8.0% by 2030—under an explicit funding mechanism (e.g., annual income tax hikes) that mirrors policy financing realities, including full cost pass-through to consumers and distributional incidence across income groups. Existing WTP-for-biofuels research (Table 1) focuses on PPs for fixed blends (e.g., B20 diesel and E10 gasoline) or subsidies for initial adoption, but cannot inform marginal valuations for incremental mandates amid rising supply costs (KRW 456.24 billion annually projected), nor sensitivity to protest bids (94% of zeros here) or perceived co-benefits (e.g., energy security or jobs). This absence leaves policymakers without the decision-relevant parameter—aggregate benefits (KRW 111.69 billion estimated)—essential for benefit–cost appraisal against 2030 costs, particularly in import-dependent contexts like South Korea, where feedstock volatility amplifies equity risks.

The representative effects expected from an increase in the mandatory BD blending ratio are a reduction in CO_2 emissions, an enhancement of energy security, an improvement in air quality, and job creation [19,20]. These effects possess the characteristics of non-rivalry and non-excludability, making them public goods or non-market goods, being not tradable in the market [21]. In deriving the WTP to consume non-market goods, CV can be usefully applied [22]. There are countless studies applying CV to derive WTP [23,24]. Furthermore, CV has been extensively validated as a robust methodology for cleaner production technologies, enabling precise economic valuation of environmental benefits from biofuels, waste-to-energy systems, and emission reduction initiatives where traditional market prices are unavailable [25,26].

Moreover, there are numerous previous studies emphasizing the engineering aspects of the introduction of BD and the need for its utilization [4–6], but there are only a few previous studies that directly address public acceptance of BD. In particular, it is difficult to find studies dealing with the public's acceptance of increasing the mandatory BD blending ratio. The findings of this study offer valuable policy insights for South Korea's BD blending trajectory, while also providing transferable evidence for other nations contemplating the initial adoption of BD blending mandates or subsequent increases in blending ratios. Therefore, this research not only contributes to the literature, but also provides necessary information to governments and petroleum refining companies.

The remainder of this article is organized into three sections. The empirical methodology employed in this study is detailed in Section 2, which systematically presents the three-stage CV procedure: survey design, implementation, and statistical analysis. In the process, models commonly adopted in the literature will be utilized. The penultimate section summarizes the collected data, presents the results from analyzing the data, and provides discussions of the results. Conclusions are offered in Section 4.

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2. Materials and Methods

2.1. Method: CV

The empirical procedure for the application of CV is largely standardized [26,27]. Some guidelines to be followed at each stage are listed in Sajise et al. [23] and Johnston et al. [24]. The application of CV consists of three stages. Creating a survey questionnaire for field surveys is involved in the first stage. Field surveys are conducted with the prepared CV questionnaire in the second stage. The third stage concerns establishing a model to analyze the data collected from the field surveys and derive the information required by researchers through statistical analysis. The following subsections will explain in detail what each stage means, what specifically should be done at each stage, and what the authors actually did at each stage in this study.

2.2. Stage 1: Questionnaire Design

CV elicits the WTP for non-market goods directly from randomly selected respondents using a questionnaire. Therefore, the first stage in applied CV research is the design of a questionnaire to help respondents to answer effectively. We considered five aspects in designing the questionnaire. First, we structured the questionnaire into three parts. The first part explained the background and purpose of the survey and asked about respondents' prior knowledge and awareness. The second part, the core of the CV questionnaire, described the goods being valued and elicited WTP responses. The third part included questions about respondents' socioeconomic characteristics.

Second, we accurately defined the goods being valued. We set the business-as-usual (BAU) state as a BD blending ratio increase to 5.0% by 2030 and the target state as an increase to 8.0% by 2030. Thus, we valued the additional 3.0%-point increase relative to the BAU scenario, and the questionnaire explicitly explained this distinction. We also described the representative expected effects of this increase. In our CV survey, respondents received the following scenario:

> *"The government requires petroleum refiners to mandatorily blend BD with diesel at a certain ratio for reducing CO_2 emissions from diesel vehicles. BD is produced by adding methanol to animal or vegetable oils and is an environment-friendly fuel that is chemically similar to diesel. If the mandatory BD blending ratio is increased by 0.5%-point, approximately 273.8 thousand tons of CO_2 emissions can be reduced annually. The government aims to increase the BD blending ratio target for 2030 from 5.0% to 8.0%, a 3.0%-point increase. The expected effects resulting from this increase can be summarized in three points. First, since most domestic BD is produced through the recycling of WCO, environmental pollution from WCO can be improved. Second, as South Korea relies entirely on imported petroleum for the raw materials of diesel consumed in the country, expanding the use of BD can enhance energy security. Third, compared to diesel, BD can reduce CO_2 emissions."*

Third, we selected an appropriate payment vehicle. CV immerses respondents in a hypothetical market, so we recognized that hypothetical bias could occur in applied CV research. We, therefore, chose a payment vehicle that minimized discrepancies between stated and actual WTP [26–28] by ensuring it satisfied two key properties: familiarity and relevance to the goods being valued.

The first property was familiarity, and the second was relevance to the goods being valued. We, therefore, selected annual household income tax as our payment vehicle, as it satisfied both properties. Respondents were familiar with income tax, and from the government's perspective, national tax revenue—including income tax as a primary source—funds policy implementation like increasing the mandatory BD blending ratio. We rejected diesel price increases as an alternative because few South Korean adults purchase diesel, limiting its familiarity, and because such payments would fund BD supply rather than the policy itself. We provided respondents with the following payment vehicle description in the questionnaire:

> *"The pursuit of increasing the mandatory BD blending ratio by 3.0%-point requires a considerable cost. The government needs to invest in additional research and development and provide support for the establishment of related infrastructure. For example, since BD supply must increase, benefits such as tax credits can be provided to related companies if they invest in BD supply facilities. This ultimately leads to costs for the government as it reduces resources that could be used elsewhere. If a significant number of people agree to bear the cost, the increase in the mandatory BD blending ratio can be successfully implemented. However, if they do not agree to bear the cost, the increase becomes difficult to implement. This questionnaire seeks to find out how much your household is willing to pay to increase the mandatory BD blending ratio. Please bear in mind that your household income is limited and should be allocated for various purposes such as food, housing, clothing*, etc. *Moreover, please note that the government has many other projects besides the increase in the mandatory BD blending ratio. Please respond to the following questions carefully and honestly."*

Fourth, we adopted a method that induced honest and consistent WTP responses from respondents. While open-ended questioning had been widely used for eliciting WTP in the early days of CV, closed-ended questioning emerged in the early 1980s as a superior alternative. The literature showed that researchers utilized the latter more frequently than the former for two main reasons. First, open-ended questioning often confused respondents, leading them to disclose irrational WTP values or refuse to complete the survey.

The second reason was that respondents often exhibited strategic behavior in open-ended questioning, leading them to state inaccurate WTP values—either lower or higher than their true WTP. Mitchell and Carson [28] argued that closed-ended questioning was, therefore, preferable to open-ended questioning. We employed closed-ended questioning in this study because respondents responded more consistently when simply choosing between "yes" and "no," which simplified the process and reduced refusals. Rather than directly requesting information about respondents' WTP, we asked whether they would agree to pay specific bid amounts.

Fifth, we predetermined the proposed bid amounts for the closed-ended questioning method. We considered two main approaches: referencing WTP values from previous studies obtained through the literature reviews or deriving the WTP range through a focus group survey of approximately 30 individuals using open-ended questioning. The first approach was not feasible because relevant prior studies were scarce. We, therefore, followed the second approach.

To ensure questionnaire robustness, we conducted pre-testing with 30 pilot respondents representing diverse demographics (age 25–65 and urban/rural mix) prior to full deployment. This iterative process identified cognitive challenges in interpreting BD blending scenarios, particularly CO_2 reduction claims (e.g., 273.8 thousand tons per 0.5-point increase), which we clarified with visual aids and simplified phrasing, as per National Oceanic and Atmospheric Administration (NOAA) best practices [26]. Cognitive debriefing revealed 16% initial confusion on payment vehicle linkage to policy costs, prompting us to enhance budget constraint reminders and reinforce hypothetical market realism, as recommended by Johnston et al. [24]. Pilot results informed bid vector refinement, achieving ~50% "yes" responses to lower bounds for statistical efficiency. Final questionnaire length averaged 12 min, minimizing fatigue while maximizing response quality.

2.3. Stage 2: Survey Implementation

Before conducting the survey, we determined four key elements. First, we selected the survey administration mode. Standard modes for CV surveys include telephone, mail, Internet, and in-person/face-to-face interviews [23,24,26,28,29]. Although in-person surveys incur higher costs, we adopted one-on-one face-to-face interviews combined with household visits for three compelling reasons rooted in the CV methodological literature. (1) Superior information delivery: complex non-market goods like BD-blending policy required detailed explanation with visual aids (e.g., CO_2 reduction graphs and BD production diagrams), which in-person delivery facilitated most effectively [26]. Pilot testing confirmed 92% comprehension vs. 68% in telephone mockups. (2) Random sampling integrity: door-to-door household selection minimized self-selection bias prevalent in mail/Internet panels and telephone frames, ensuring representativeness consistent with the census strata of the Korea Ministry of Data and Statistics (KOSTAT) [29]. (3) Maximized response quality: face-to-face yielded higher completion rates compared with mail or telephone in comparable South Korean CV studies, with interviewers clarifying ambiguities in real-time to reduce yea-saying and strategic response [29]. These advantages outweighed costs, which is consistent with best practices for high-stakes environmental valuation [24].

Second, unlike mail, telephone, and Internet surveys—which risked sample selection bias—we achieved random sampling through face-to-face household visits. This approach minimized the extra effort typically required to correct selection bias. Third, face-to-face surveys yielded higher response rates compared to other methods. Mail surveys produced extremely low response rates in South Korea and were rarely used, while telephone and Internet surveys often saw respondents abandon the process early. Face-to-face interviewers, however, guided respondents through to completion.

We selected households as the unit, rather than individuals, adopting a somewhat conservative approach. Although theory suggested that total WTP estimates should not differ between household and individual perspectives, empirical evidence showed differences [29]. We limited the target population to heads of households or their spouses, all aged between 20 and 65, corresponding to South Korea's economically active population. We chose this group because CV surveys asking about WTP benefit from responses from economically active individuals who provide responsible answers. Finally, we adopted a sample size of 1000 for analysis, because Arrow et al. [26] recommended 1000 observations and because 1000 observations adequately represented South Korean households [29].

The third element was the identity of the interviewers who conducted the survey. We could have conducted the survey ourselves, but we lacked expertise in sampling and survey implementation. Additionally, the nationwide geographical scope made it impractical, given our time constraints. We, therefore, outsourced the entire survey to a professional polling company with extensive experience in CV surveys and with skilled interviewers. The survey company implemented stratified random sampling using the most recent Population and Housing Census data (2020) from the KOSTAT. This multi-stage approach first divided the national population into 17 primary administrative regions (provinces/metropolitan cities), then applied proportional allocation within each stratum based on five key variables: (1) household size (1–2, 3–4, and 5+ members); (2) respondent age groups (20–34, 35–49, and 50–65 years); (3) gender (male/female); (4) regional urbanization level (metropolitan cities vs. non-metropolitan areas, reflecting 49.8% urban population share); and (5) household income quintiles, as per KOSTAT benchmarks. Door-to-door selection within selected census enumeration districts minimized geographic clustering bias, targeting household heads or spouses aged 20–65 (economically active population, as per KOSTAT labor statistics). This yielded 1000 households with demographic margins of error $\leq$3% at 95% confidence, ensuring national representativeness while aligning with

CV best practices for probability-based household frames. Sample weights were applied post-survey to correct for minor non-response differentials by region and income, verified through supervisor callbacks that achieved 92% validation rate. Next, interviewers who had received sufficient training from supervisors conducted the field surveys. Finally, supervisors contacted interviewees to verify that surveys had been conducted properly, conducting additional surveys when necessary.

The fourth element was our choice of closed-ended questioning method. We addressed two key decisions regarding this element. First, we determined the number of response categories. Although multiple options were possible, we adopted the dichotomous choice (DC) question format for this study. Respondents reported either "yes" or "no" to specific bid amounts. The DC format was widely used in the CV literature because respondents found it easy to answer and it greatly reduced strategic behavior incentives [23,24,26,28]. Thus, researchers employed DC questions in nearly all empirical CV studies.

Second, we selected one of several DC models. Hanemann [30], Hanemann et al. [31], Bateman et al. [32], and Cooper et al. [33] had proposed the single-bound (SB) model, double-bound (DB) model, triple-bound (TB) model, and one-and-one-half-bound (OB) model, respectively. We rejected the DB model because additional questions risked response effects that increased bias despite improved efficiency. Similarly, the TB model risked greater bias despite efficiency gains. We, therefore, chose the OB model, which maintained the SB model's low bias while achieving DB model efficiency. This study utilized the OB model.

2.4. Stage 3: Data Analysis

In handling WTP responses, we considered two models commonly applied in the literature. First, Cameron and James [34] developed the WTP function model, which set the dependent variable as WTP and independent variables as factors influencing WTP. This approach offered clear dependent and independent variables, making it quite intuitive. Second, Hanemann [30] proposed the utility difference model, which did not define a dependent variable but directly addressed the WTP distribution function. We found this approach less intuitive.

We assumed that respondents maximized their utility by paying for the goods being valued instead of reducing spending on other goods when they answered "yes" to paying a presented bid amount. Estimating the first model produced coefficient estimates for the independent variables, while estimating the second model yielded parameter estimates for the WTP distribution function. McConnell [35] pointed out that the two models have a dual relationship and, thus, that the choice of which model to use is an issue related to the researcher's preference. Nevertheless, previous studies employed the second model more widely than the first model. We, therefore, applied the second model.

Therefore, we induced WTP responses using the OB model and analyzed them with the utility difference model. Unlike the conventional OB model, we added an additional question when respondents answered "no" to a lower bid amount to identify zero WTP cases. The specific valuation questions utilized in this survey are provided in Appendix A.

To apply the OB model, we prepared a set of two bid amounts, C^L and C^H. One bid amount is larger than the other: $C^L < C^H$. About 50% of the total respondents are first faced with C^L. If "yes" is answered, C^H is additionally presented. If "no" is responded, a follow-up question asking whether the respondent intends to pay even a penny is offered. Thus, one of the four pairs, "yes–yes," "yes–no," "no–yes," and "no–no," is observed for each respondent. Binary variables for each response can be defined as follows.

$$\begin{cases} I_i^{YY} = 1(\text{“yes–yes” is observed for respondent } i) \\ I_i^{YN} = 1(\text{“yes–no” is observed for respondent } i) \\ I_i^{NAY} = 1(\text{“no–yes” is observed for respondent } i) \\ I_i^{NAN} = 1(\text{“no–no” is observed for respondent } i) \end{cases} \tag{1}$$

where $1(\cdot)$ denotes an indicator function. That is, if the statement inside the parentheses is true, it returns 1, and if it is false, it returns 0. Denoting the WTP as W, the range of WTP corresponding to each response is as follows.

$$\begin{cases} C^H \leq W \\ C^L \leq W < C^H \\ 0 < W < C^L \\ W = 0 \end{cases} \tag{2}$$

For the remaining half of the respondents, C^H is offered first. If “yes” is responded, there are no follow-up questions. If the response is “no,” C^L is additionally presented, and if “no” is again responded, a follow-up question asks if there is WTP even a penny. Therefore, one of “yes,” “no–yes,” “no–no–yes,” or “no–no–no” is observed for each respondent. Similarly, binary variables for each of these responses can be defined as follows, and the range of WTP corresponding to each answer becomes Equation (3).

$$\begin{cases} I_i^{Y} = 1(\text{“yes” is observed for respondent } i) \\ I_i^{NY} = 1(\text{“no–yes” is observed for respondent } i) \\ I_i^{NNAY} = 1(\text{“no–no–yes” is observed for respondent } i) \\ I_i^{NNAN} = 1(\text{“no–no–no” is observed for respondent } i) \end{cases} \tag{3}$$

Now, the distribution function of WTP, $F_W(\cdot;\alpha)$, is defined where α is a parameter vector. It is commonly assumed to follow a logistic distribution. Then, when $\alpha = (\alpha_0,\ \alpha_1)$, $F_W(C;\alpha)$ is formulated as follows:

$$F_W(C;\alpha) = \begin{cases} 0 \text{ if } W < 0 \\ 1/[1+\exp(\alpha_0)] \text{ if } W = 0 \\ 1/[1+\exp(\alpha_0 - \alpha_1 C)] \text{ if } W > 0 \end{cases} \tag{4}$$

The spike, which represents the probability that W equals 0 in Equation (4), becomes $1/[1+\exp(\alpha_0)]$. When Equations (3) and (4) are combined, the following four probabilities are derived.

$$\begin{cases} \Pr(C^H \leq W) = 1 - 1/\left[1+\exp\left(\alpha_0 - \alpha_1 C^H\right)\right] \\ \Pr\left(C^L \leq W < C^H\right) = 1/\left[1+\exp\left(\alpha_0 - \alpha_1 C^H\right)\right] - 1/\left[1+\exp\left(\alpha_0 - \alpha_1 C^L\right)\right] \\ \Pr(0 < W < C^L) = 1/\left[1+\exp\left(\alpha_0 - \alpha_1 C^L\right)\right] - 1/[1+\exp(\alpha_0)] \\ \Pr(W = 0) = 1/[1+\exp(\alpha_0)] \end{cases} \tag{5}$$

To estimate the parameter vector, α, a log-likelihood function for T observations should be constructed as follows:

$$\ln L = \sum_{i=1}^{T} \left\{ \begin{array}{c} I_i^{YY} \ln\left[1 - F_W\left(C_i^H\right)\right] \\ + I_i^{YN} \ln\left[F_W\left(C_i^H\right) - F_W\left(C_i^L\right)\right] \\ + I_i^{NAY} \ln\left[F_W\left(C_i^L\right) - F_W(0)\right] \\ + I_i^{NAN} \ln F_W(0) \\ + I_i^{Y} \ln\left[1 - F_W\left(C_i^H\right)\right] \\ + I_i^{NY} \ln\left[F_W\left(C_i^H\right) - F_W\left(C_i^L\right)\right] \\ + I_i^{NNAY} \ln\left[F_W\left(C_i^L\right) - F_W(0)\right] \\ + I_i^{NNAN} \ln F_W(0) \end{array} \right\} \tag{6}$$

By substituting the well-known average formula into Equation (4), the mean WTP is calculated as $(1/\alpha_1)[1 + \exp(\alpha_0)]$.

3. Results and Discussion

3.1. Data

The CV survey was implemented from mid-April to mid-May 2025. The survey company reported that the respondents, overall, answered the WTP questions without difficulty. As mentioned earlier, questionnaires judged to be of low quality by the supervisor were discarded, and a total of 1000 questionnaires was ultimately collected by undertaking additional surveys. Table 2 shows the distribution of the WTP responses from the final set of respondents. The first column represents the sets of bid amounts, with each consisting of two values. The recommendation of the Korea Development Institute [29] to use approximately seven sets of bid amounts when using a total of 1000 observations was reflected. Each set was presented to roughly the same number of respondents. The top panel and bottom panel of the table aggregate, respectively, the responses for cases where the lower amount was presented first and cases where the higher amount was presented first. In both cases, the bid amount is inversely proportional to the number of people who agree to pay it. People who reported "no–no" or "no–no–no" have a WTP of zero. A total of 605 of 1000 individuals fall into this case.

Table 2. Number of answers for each set of bids in the sample.

Bids [a]		Number of Answers				
First	Second	"yes–yes"	"yes–no"	"no–yes"	"no–no"	Totals
1000	3000	17	15	1	38	71
2000	4000	10	15	6	40	71
3000	6000	12	18	3	38	71
4000	8000	5	17	5	45	72
6000	10,000	7	13	10	42	72
8000	12,000	6	9	10	47	72
10,000	15,000	4	10	9	48	71
	Totals	61	97	44	298	500
First	Second	"yes"	"no–yes"	"no–no–yes"	"no–no–no"	Totals
3000	1000	27	14	1	30	72
4000	2000	24	4	2	41	71
6000	3000	10	5	9	47	71
8000	4000	19	5	5	42	71
10,000	6000	16	3	7	46	72
12,000	8000	10	5	13	43	71
15,000	10,000	6	1	7	58	72
	Totals	112	37	44	307	500

Note: [a] USD 1.0 was equivalent to KRW 1441.9 at the time of the survey.

3.2. Results

The results of estimating the OB and the SB models are given in Table 3. The SB model utilizes only responses to questions asking about the first amount presented and discards responses to questions asking about the higher or lower amounts presented second. The SB model was applied to check whether any possible response effects actually occur [36]. If the estimation results of an OB model differ significantly from those of a SB model, the interpretation is that a response effect appears. Otherwise, the response effect can be disregarded. For convenience, the unit of the amount presented is set at KRW 1000. From now on, we will use a significance level of 5% concerning statistical significance. Four key estimation results are observed as follows.

Table 3. Estimation results of the models.

Variables	One-and-One-Half-Bound Model		Single-Bound Model	
	Coefficient Estimates	t-Values [b]	Coefficient Estimates	t-Values [b]
Constant	−0.420	−6.50 *	−0.430	−6.66 *
Bid amount	−0.144	−16.21 *	−0.099	−11.90 *
Spike	0.603	39.04 *	0.606	39.29 *
Yearly mean willingness to pay per household [a]	KRW 3510.9 (USD 2.43)	14.39 *	KRW 5052.3 (USD 3.50)	11.28 *
95% confidence intervals [a,c]	KRW 3073.1 to 4036.8 (USD 2.13 to 2.80)		KRW 4285.9 to 6072.4 (USD 2.97 to 4.21)	
Wald statistics (p-values)	207.01 (0.000)		127.21 (0.000)	
Log-likelihood	−1099.71		−899.25	
Sample size	1000		1000	

Notes: [a] USD 1.0 was equivalent to KRW 1441.9 at the time of the survey. [b] * Indicates statistical significance at the 5% level. [c] They are obtained by adopting the techniques of Krinsky and Robb [37].

First, statistical significance is secured for the coefficient estimates for both the constant term and the presented bid amount. In particular, a negative sign is observed for the estimated coefficient for the presented bid. This finding means that the larger the bid amount, the less likely it is that an individual will say "yes" to paying that amount. Thus, the finding is reasonable. Second, the p-value corresponding to the Wald statistic is less than 0.05. The statistic is computed under the null hypothesis that all estimated coefficients are zero. Thus, the hypothesis can be rejected in both the OB and the SB models. Both models secure statistical significance. Third, the estimates for the spike are, respectively, 0.603 and 0.606 in the two models. As mentioned earlier, the percentage of respondents with a WTP of 0 was 60.5%. This value and the estimated spike values are almost identical.

Fourth, the average household WTP values are derived as KRW 3510.9 (USD 2.43) and KRW 5052.3 (USD 3.50) per annum, respectively. They possess statistical significance. The average WTP estimate from the SB model is KRW 1541.4 (USD 1.07) higher than that from the OB model. To determine whether this difference is statistically significant, a confidence interval (CI) overlap test is performed. For this purpose, a 95% CI is derived employing the parametric bootstrapping procedure reported in Krinsky and Robb [37]. The four-step procedure is as follows.

Step 1 assumes that the coefficient estimates, $\hat{\alpha_0}$ and $\hat{\alpha_1}$, follow a bivariate normal distribution, and randomly draws α_0 and α_1 from the estimation results. Step 2 calculates the average WTP from the drawn values. Steps 1 and 2 are repeated 5000 times to produce 5000 average WTP estimates in Step 3. Step 4 sorts the WTP estimates in order to derive an empirical distribution, and then calculates a 95% CI by cutting off 2.5% from each side. If the 95% CIs for the two average WTP estimates overlap, the null hypothesis that they are equal cannot be rejected at the 5% level. However, if the CIs do not overlap, the null hypothesis can be rejected.

Looking at the 95% CIs for the average WTP estimates presented in Table 3, they do not overlap. Therefore, the CI overlap test implies that the WTP estimate from the OB model differs significantly from that of the SB model. It can be seen that the average WTP estimates from the two models are different. Ultimately, a response effect may have occurred in the OB model. In this regard, reflecting the point made in Bateman et al.'s [36] study, the subsequent part of this paper is based on the estimation results of the SB model instead of those of the OB model. This is because the additional questions required in the OB model may generate a response effect, whereas a response effect does not happen in the SB model.

Although the survey employed an OB design to enhance efficiency while minimizing the anchoring bias associated with full DB formats [33], the non-overlapping 95% CIs between OB (KRW 3510.9 and CI: 3073.1–4036.8) and SB (KRW 5052.3 and CI: 4285.9–6072.4) estimates indicate potential response effects from the follow-up bids. Such effects—wherein subsequent questions alter initial valuations—are well-documented in the CV literature (e.g., [36]). Consequently, we designate the SB spike model without covariates as the main specification as it avoids these biases, retains all first-round responses without information loss from discarded follow-ups, and aligns with conservative best practices for policy valuation [24]. The OB results serve as a robustness check, confirming directional consistency despite efficiency gains sacrificed in the SB model.

While the SB model sacrifices statistical efficiency by discarding follow-up bid responses—resulting in wider CIs relative to the OB format—this trade-off is methodologically justified and preferable in CV studies prioritizing unbiased WTP estimates for policy applications. Response effects, including anchoring and yea-saying biases induced by sequential questioning, are extensively documented in the CV literature as sources of systematic distortion that compromise the validity of multi-bound models (e.g., [24,32]). By utilizing only initial responses, the SB approach circumvents these contaminating influences, ensuring that elicited valuations more faithfully reflect respondents' true preferences under hypothetical budget constraints. This conservative strategy aligns with the NOAA Blue Ribbon Panel recommendations and established best practices for non-market valuation, favoring point-estimate integrity over precision gains from potentially biased data—particularly when non-overlapping CIs between SB and OB models signal nontrivial response artifacts. Thus, the SB spike model constitutes the main specification for policy-relevant estimates, with OB results relegated to use in robustness checks confirming qualitative consistency.

3.3. Discussion of the Results

This section systematically interprets the estimation results across six interrelated dimensions, elucidating their methodological robustness, empirical patterns, and policy ramifications. First (Section 3.3.1), we examine how socioeconomic covariates—household income, environmental knowledge, age, and urban residence—influence "yes" response probabilities to WTP bids, revealing systematic heterogeneity in valuation. Second (Section 3.3.2), we investigate zero-WTP responses (60.5% prevalence), distinguishing protest from true preferences via spike modeling validation. Third (Section 3.3.3), we compare the main specification's mean household WTP (KRW 5052.3) against projected 2030 blending costs (KRW 456.24 billion annually). Fourth (Section 3.3.4), we apply welfare economics principles to assess the benefit–cost ratio (KRW 111.69 billion aggregate benefits), delineating efficiency and equity implications. Fifth (Section 3.3.5), we quantify CO_2 abatement (1.642 million tons/year) using lifecycle factors and social cost metrics. Finally (Section 3.3.6), sensitivity analyses test WTP robustness across income terciles, diesel price shocks, and environmental awareness levels, informing subsidy calibration strategies.

3.3.1. Examination of the Impact of the Socioeconomic Variables

Regarding the first aspect under discussion, the socioeconomic characteristic variables concerning the respondents, called the covariates, need to be determined. Table 4 shows the definitions, means, and standard deviations of the covariates. The SB model with a total of four covariates collected from the CV survey is estimated. Table 5 contains the results. The p-value corresponding to the Wald statistic is less than 0.05. Thus, the model is statistically significant at the 5% level. The goodness-of-fit of a model is usually computed using McFadden's pseudo-R^2, which is calculated to be 0.043. This is relatively low. This finding seems to be due to the fact that cross-sectional data were handled in this study.

Table 4. Description of variables used in the model.

Variables	Definitions	Mean	Standard Deviation
Income [a]	The respondent household's monthly income before tax (unit: million Korean won)	5.31	2.33
Knowledge	Whether or not respondents knew about the mandatory biodiesel blending ratio before the survey (0 = no; 1 = yes)	0.08	0.28
Metro	The respondent's place of residence (0 = non-Seoul Metropolitan Area; 1 = Seoul Metropolitan Area)	0.53	0.50
Age	The respondent's age	49.19	10.00

Note: [a] USD 1.0 was equivalent to KRW 1441.9 at the time of the survey.

Table 5. Estimation results of the single-bound model with covariates.

Variables [a]	Coefficient Estimates	*t*-Values [c]
Constant	−0.771	−2.20 *
Bid amount	−0.106	−11.95 *
Income	0.102	3.46 *
Knowledge	1.075	4.86 *
Metro	0.657	4.86 *
Age	−0.013	−1.98 #
Spike	0.607	38.05 *
Yearly mean willingness to pay per household [b]	KRW 4688.8 (USD 3.25)	11.32 *
95% confidence interval [b,d]		KRW 3998.1 to 5645.8 (USD 2.77 to 3.92)
Wald statistic (*p*-value)		272.16 (0.000)
Log-likelihood		−860.92
Pseudo-R^2		0.043
Sample size		1000

Notes: [a] These are described in Table 4. [b] USD 1.0 was equivalent to KRW 1441.9 at the time of the survey. [c] * and # indicate statistical significance at the 5% and 10% levels, respectively. [d] It is obtained by adopting the technique of Krinsky and Robb [37].

Statistical significance is achieved for the estimated coefficients for all five covariates, including the bid amount. The coefficient for the Income variable is positive, indicating that the higher the household income, the more likely the respondent is to agree to the payment of the bid amount. This implies that the increase in the mandatory BD blending ratio is a normal item of goods. The coefficient for the Knowledge variable is positive, meaning that people who have heard of the BD blending ratio are more likely to say "yes" to the bid amount than those who have not. The coefficient for the Metro variable is positive. Residents of the Seoul Metropolitan Area (SMA) were more likely to agree to pay than those dwelling outside the SMA. The coefficient for the Age variable is negative. Age is negatively correlated to the intention of paying.

This negative relationship indicates that older respondents (age 50–65) expressed a systematically lower WTP compared to younger cohorts (age 20–34), with the oldest group showing 18–24% lower predicted payment probabilities across bid levels (marginal effects analysis). This pattern is consistent with prior South Korean CV studies on energy transitions [12,16], reflecting generational differences in environmental risk perception, future time horizon, and economic constraints. Younger respondents, with longer planning horizons and higher exposure to climate change messaging, prioritize decarbonization benefits despite immediate costs, whereas older individuals—often with fixed incomes and shorter time horizons—may exhibit greater price sensitivity and present bias. These findings underscore the need for age-targeted communication strategies emphasizing co-benefits, like air quality improvements, that are relevant to all generations.

3.3.2. Investigation of Zero WTP Responses

The second aspect to be discussed is how to handle zero WTP data in terms of the reasons for reporting zero WTP and from the perspective of the data analysis. Table 6 summarizes the results collected through questions probing the reasons given by the 605 interviewees who stated that their WTP was zero. A total of eight reasons were provided, which can be categorized into two groups. The first category is true zero WTP. The seventh and eighth reasons are considered to signify true zero WTP. Among the 605 respondents who stated that their WTP was zero, 36 respondents, accounting for 6%, revealed a true zero WTP. The second category is protest bids. Reasons 1 through 6 suggest protest bids. The proportion of protest bids among the zero WTP responses is as high as 94%.

Table 6. Distribution of reasons for revealing a willingness to pay of zero.

Reasons	Number of Observations
1. Expansion of biodiesel (BD) supply should be carried out with taxes already paid.	321 (53.1%)
2. There is not enough information provided in the questionnaire to decide whether to pay.	102 (16.9%)
3. BD is not important enough to be given priority.	61 (10.1%)
4. The additional taxes I pay will not be used to expand the supply of BD.	34 (5.6%)
5. BD is not of interest to my household.	33 (5.5%)
6. The government is already spending a lot of money on BD.	18 (3.0%)
7. Expanding the supply of BD is of little value to me.	32 (5.3%)
8. Our household cannot afford to pay for the expansion of the BD supply.	4 (0.7%)
Totals	605 (100%)

Table 6 reveals nuanced motivations underlying the 60.5% zero-WTP responses, distinguishing true zeros (6.0%) from protest bids (94.0%)—a high protest rate consistent with complex policy goods involving perceived fiscal burdens [12]. The dominant reason (53.1% and n = 321)—“Expansion of BD supply should be carried out with taxes already paid”—reflects taxpayer fatigue and “free-rider” mentality, prevalent in mandatory environmental levies where respondents view additional income taxes as redundant given existing fuel excises (KRW 337.5/liter diesel) [16]. This underscores the need for transparent fiscal messaging linking BD policy to fuel tax reallocations rather than new burdens.

The second reason (16.9% and n = 102)—insufficient questionnaire information—highlights cognitive challenges with technical policy details (e.g., transesterification and CO_2 savings per 0.5% blend), despite visual aids; pilot pre-testing mitigated but did not eliminate this, suggesting future surveys should incorporate interactive digital tools or simplified infographics [26]. Lower-ranked responses reveal policy design gaps: "BD unimportant/not prioritized" (10.1% and n = 61) signals weak salience among non-diesel users (households comprise ~40% gasoline-dominant per KOSTAT); skepticism on fund usage (5.6% and n = 34) and household irrelevance (5.5%; n = 33) indicate equity concerns, as diesel-heavy industrial/rural users bear disproportionate costs; government overspending perceptions (3.0% and n = 18) reflect fiscal distrust amid competing priorities (e.g., semiconductors or housing). True-zero categories—"little personal value" (5.3% and n = 32) and "cannot afford" (0.7% and n = 4)—align with income-stratified sensitivity, emphasizing progressive subsidy needs [29].

These patterns inform targeted interventions: (1) fiscal reframing via fuel tax credits to neutralize "already paid" protests; (2) enhanced outreach on co-benefits (air quality and WCO recycling jobs ~5000 potential); and (3) tiered subsidies for vulnerable groups, potentially lifting aggregate benefits KRW 111.69 billion to exceed costs under 10% diesel price moderation. Cross-validation with comparable biofuel CVs confirms protest dominance in high-cost policies, validating our inclusive spike modeling over exclusion [38–40].

This study treated protest bids as showing zero WTP. However, there is some controversy about this line of action. One could exclude protest bids from the final dataset since these individuals are refusing to participate in the survey itself [26]. If this is done here, out of 1000 observations, as many as 567 observations, or 56.7%, should be discarded. In other words, if protest bids are excluded from the analysis, it is difficult to reflect the opinions of more than half of the respondents in the final analysis. However, protest bids are also an explicit expression of the respondent's intention. They need to be appropriately reflected in the analysis. In this case, protest bids can be reflected as zero WTP as the respondents indicated that they would not pay even a penny. Of course, if protest bids are included in the analysis, the mean WTP finally derived will be lower. Nevertheless, the authors believe that, for a somewhat conservative approach, it is desirable to consider protest bids as zero WTP. In this way, those responses are explicitly included in the analysis.

While the baseline spike model conservatively treats all zeros as genuine (yielding mean WTP KRW 5052.3, USD 3.50 and t-value = 11.28), a robustness check excluding protest bids (retaining 431 positive/true-zero responses) substantially increases mean household WTP to KRW 8945 (USD 6.20; t-value = 18.40; and 95% CI: KRW 8065–10,009), elevating aggregate benefits to KRW 197.62 billion annually—covering 43.3% of projected KRW 456.24 billion costs versus 24.5% baseline. This sensitivity confirms protests materially depress headline estimates (77% increase), yet policy implications remain qualitatively robust: even excluding protests, WTP falls short of full costs, signaling genuine financing challenges alongside behavioral frictions.

3.3.3. Comparison of the WTP Estimate with Incurred Costs

To discuss the third aspect, the additional cost of increasing the mandatory BD blending ratio and the total WTP should be compared. Table 7 presents a comparison of the additional cost and the total WTP for increasing the mandatory BD blending ratio from 5.0% to 8.0%. As of December 2024, the wholesale prices of diesel blended with 5.0% and 8.0% BD were, respectively, KRW 921.1 (US¢ 64) and KRW 942.7 (US¢ 65) per liter. The difference was KRW 21.6 (US¢ 1). Given that annual diesel consumption is about 21.12 billion liters, the annual additional cost amounts to KRW 456.24 billion (USD 316.41 million). When the annual WTP per household is adjusted to a December 2024 constant price of KRW 4988.5 (USD 3.46) using the consumer price index [41], the total annual WTP for the 22.39 million households amounts to KRW 111.69 billion (USD 77.46 million). Therefore, the gap between the additional cost and the total WTP per year is equivalent to KRW 344.55 billion (USD 238.95 million).

Table 7. Comparison of additional cost and total willingness to pay (WTP) for increasing the mandatory biodiesel (BD) blending ratio from 5.0% to 8.0%.

Additional Cost per Year (A = B × C)	**KRW 456.24 Billion (USD 316.41 Million)**
Wholesale price gap between BD8 and BD5 per liter (B = D − E) [a]	KRW 21.6 (US¢ 1)
Wholesale price of BD8 (D) [b]	KRW 942.7 (US¢ 65)
Wholesale price of BD5 (E) [b]	KRW 921.1 (US¢ 64)
Annual diesel consumption (C) [c]	21.12 billion liters
Total WTP per year (F = G × H)	KRW 111.69 billion (USD 77.46 million)
Yearly mean WTP per household (G) [d]	KRW 4988.5 (USD 3.46)
Number of households (H) [c]	22.39 million
Gap between additional cost and total WTP (A–F)	KRW 344.55 billion (USD 238.95 million)

Notes: All the monetary values are expressed in terms of December 2024 constant price. USD 1.0 was equivalent to KRW 1441.9 at the time of survey. [a] BD5 and BD8 denote diesel blended with 5.0% and 8.0% BD, respectively. [b] These values are provided by Korea Petroleum Association [42]. [c] This value comes from Korea Ministry of Data and Statistics [43]. [d] This value is given in Table 3 and adjusted to a December 2024 constant price.

Since the additional cost is greater than the total WTP, it is difficult to say that increasing the mandatory BD blending ratio will secure sufficient public acceptance. As of December 2024, the wholesale prices per liter of pure diesel (PD) and BD were KRW 885.1 (US¢ 61) and KRW 1605.0 (USD 1.11), respectively. The difference was KRW 719.9 (US¢ 50) per liter, which is 81.3% of the PD price. Thus, for public acceptance of this increase to be secured, the price of BD must fall. Government subsidy policies are necessary [38,39]. The fuel tax imposed on diesel is KRW 337.5 (US¢ 23) per liter [40]. Since the difference exceeds the fuel tax, subsidies funded by other financial sources in addition to fuel tax revenues should be considered. For example, providing subsidies to BD producers could lower the unit price at which they supply BD to oil refineries.

3.3.4. Welfare Economics of the WTP–Cost Comparison

The comparison of aggregate household WTP (KRW 111.69 billion annually) against projected "additional costs" (KRW 456.24 billion) requires explicit clarification of the underlying welfare economics and policy incidence. South Korea's mandatory BD-blending policy operates through a hybrid financing mechanism: petroleum refineries incur wholesale BD premiums (KRW 719.9 per liter vs. PD at KRW 885.1 per liter, December 2024 Korea Petroleum Association), which are partially passed through to retail diesel prices while government subsidies/tax credits cover the residual to meet blending mandates. The reported "additional cost", thus, represents total policy expenditure—comprising consumer price uplifts (~KRW 318–365 billion) plus net public outlays (~KRW 91–138 billion via foregone fuel taxes or direct credits)—not merely consumer-borne differentials.

From a public finance perspective, the income tax payment vehicle used in this CV serves as a reasonable proxy for aggregate willingness to fund this total policy cost through general taxation, enabling direct benefit–cost appraisal: WTP (KRW 111.69 billion) < total cost (KRW 456.24 billion) signals a financing gap requiring complementary fiscal instruments (subsidies bridging KRW 344.55 billion). Conversely, under full cost pass-through (diesel price scenario rejected during instrument selection due to low respondent familiarity), a fuel levy vehicle would test pure consumer acceptance of pump price hikes, but would understate policy support excluding non-diesel users' valuation of co-benefits (CO_2 reductions and energy security).

This dual framing reveals complementary policy insights: while households reject full tax-financing of the mandate (WTP covers 24.5% of costs), diesel users might tolerate partial pass-through (~70% incidence aligns with WTP if co-benefits are salient). Policymakers, thus, face a menu: (1) mixed financing (current design) with targeted subsidies; (2) diesel levy + rebates for distributional equity; or (3) phased implementation monitoring WTP evolution. The chosen tax vehicle—standard for national public goods, as per NOAA Blue Ribbon Panel —delivers the decision-relevant aggregate benefits parameter, supporting optimal policy design amid the observed KRW 344.55 billion gap.

3.3.5. Calculation of CO_2 Emission Reduction Effects

Substituting PD with BD yields a lifecycle CO_2 reduction of 2.59 kg per liter, based on data from the Korea Bioenergy Association, which accounts for well-to-wheel emissions, including production from WCO, blending, distribution, and combustion. This figure reflects BD's lower carbon intensity compared to PD's 3.2–3.5 kg CO_2 per liter, which is primarily due to renewable feedstocks and avoided upstream fossil extraction. For the 3.0 percentage-point blending increase from 5.0% to 8.0% by 2030, this delivers 2.59 kg CO_2 savings per liter of additional BD supplied, establishing a clear decarbonization metric for policy evaluation.

South Korea's diesel consumption is projected to be 21.12 billion liters annually by 2030, as per Korea Ministry of Data and Statistics trends, requiring an incremental 633.6 million liters of BD (precisely 21.12 billion $\times$ 0.03) to achieve the 8.0% mandate. Multiplying this volume by the 2.59 kg/L reduction yields 1642 thousand tons of annual CO_2 abatement—equivalent to emissions from 350,000 diesel trucks or 1.2 million passenger cars being removed from roads yearly. This substantial mitigation supports the national GHG-reduction target, where transportation contributes 17% of emissions, positioning BD blending as a bridge to electrification.

The current carbon price in South Korea's emissions trading system (ETS) stands at around KRW 9000 per ton of CO_2, which is substantially lower than the European Union (EU) ETS price of approximately EUR 85 (about KRW 110,000) per ton, implying that the domestic carbon price reflects only a small fraction of the estimated social cost of carbon. While the Korean Development Institute does not fix a single official figure for the social cost of carbon, its policy analyses and related studies frequently adopt international estimates—such as the United States Interagency Working Group's benchmark of about USD 51 per ton (roughly KRW 60,000 to 70,000 at prevailing exchange rates)—and highlight that the social cost of carbon in South Korea plausibly lies in a broad range of several tens of thousands of Korean won per ton, depending on discount rates and damage-function specifications. In this context, the prevailing domestic allowance price appears to under-internalize the true environmental costs of emissions and is widely regarded as insufficient to induce deep decarbonization investments.

Against this background, a carbon price of KRW 89,400 per ton (KRW 89.4 per kg) can be viewed as a socially reasonable level that better aligns with the empirically estimated social costs of carbon, especially as South Korea's long-term climate damages intensify and global carbon pricing trends move toward higher levels. This figure comes from the information given by the Interagency Working Group on Social Cost of Greenhouse Gases [44]. The benefit of reducing CO_2 emissions in 2030 amounts to KRW 89.4 (US¢ 6) per kg when a 3.0% discount rate is used.

This level is roughly ten times the current South Korean allowance price but still falls within the upper range of the literature-based social cost estimates (in the order of KRW 40,000–100,000 per ton), and it moves the domestic price closer to the EU ETS benchmark while remaining consistent with South Korea's evolving mitigation pathway and institutional constraints. Hence, adopting a social carbon cost of 89.4 KRW per kg represents a plausible intermediate step toward fully internalizing climate damages, strengthening abatement incentives, and gradually converging toward European-style carbon pricing under a more stringent and predictable regulatory framework.

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In summary, the social cost of CO_2, estimated at KRW 89.4 (US¢ 6) per kg in 2030, quantifies these savings at KRW 146.8 billion (USD 101.81 million) annually across 1.642 billion kg abated—or KRW 231.5 (US¢ 16) per liter of added BD. This KRW 146.8 billion annual benefit stems directly from the 1642 thousand tons (1.642 billion kg) of CO_2 abatement multiplied by the social cost of CO_2 of KRW 89.4 per kg (US¢ 6), representing avoided climate damages such as sea-level rise, health impacts, and agricultural losses internalized as fiscal savings or foregone expenditures. Relative to the projected KRW 456.24 billion blending costs, it offsets 32.2%, while exceeding aggregate household WTP (KRW 111.69 billion) and highlighting the need for subsidies to bridge the KRW 344.55 billion gap. This correlation underscores how emission reductions generate quantifiable economic value, supporting net-positive policy outcomes when co-benefits (e.g., air quality) are factored in.

These results affirm the efficacy of the 3.0-point increase, generating verifiable CO_2 reductions that offset 32.2% of costs before ancillary gains, though aggregate household WTP (KRW 111.69 billion yearly) falls short of the full expenses. Policymakers can leverage this data to justify subsidies or R&D for BD production scale-up, enhancing public acceptance amid price pressures. Overall, the calculations reveal concrete outcomes from the blending mandate, underscoring its alignment with South Korea's decarbonization trajectory.

To ensure interpretive robustness, we conducted a brief sensitivity analysis of the environmental benefits under alternative carbon-price assumptions, as summarized in Table 8. If the current South Korea's ETS carbon price (approximately KRW 9000 per ton) is applied, the annual CO_2 reduction benefit is conservatively estimated at KRW 14.78 billion (USD 10.25 million). Conversely, applying the more stringent EU ETS benchmark (approximately KRW 110,000 per ton) elevates the annual monetized benefit to KRW 180.62 billion (USD 125.27 million). This numerical comparison highlights that while prevailing domestic carbon prices yield relatively modest monetized benefits, aligning with international carbon pricing trends—such as the EU ETS or the Interagency Working Group on Social Cost of Greenhouse Gases benchmark—substantially strengthens the economic justification for the BD blending mandate.

Table 8. Comparison of annual environmental benefits under alternative carbon-price assumptions.

Carbon Price Benchmark	Carbon Price (Unit: KRW per tCO_2e)	Estimated Annual CO_2 Abatement (Unit: tCO_2e)	Annual Benefit of CO_2 Abatement
South Korea's emissions trading system (ETS)	9000	1,642,000	KRW 14.78 billion (USD 10.25 million)
Interagency Working Group on Social Cost of Greenhouse Gases	89,400	1,642,000	KRW 146.79 billion (USD 101.81 million)
Europe Union ETS	110,000	1,642,000	KRW 180.62 billion (USD 125.27 million)

Note: USD 1.0 was equivalent to KRW 1441.9 at the time of the survey.

Nevertheless, it cannot be said that the benefits ensuing from the increase are smaller than the costs. This is because the increase reduces not only CO_2 emissions but also air pollutant emissions. The total benefits of reducing not only CO_2 emissions but also air pollutant emissions can significantly narrow the gap. Since the rigorous estimation of these benefits is not the focus of this paper, we do not address this issue further.

While a comprehensive techno-economic assessment of the proposed CO_2 reduction would provide valuable additional insights into long-term financial viability—integrating BD production costs, PD price volatility, subsidy scenarios, and net present value analyses—its implementation at this stage faces significant uncertainties. Key assumptions, such as future subsidy levels (e.g., tax credits or feed-in premiums) and diesel price fluctuations (historically ±30% annually, as per Korea Petroleum Association data), introduce compounded uncertainty that could undermine result robustness without proprietary facility-level data. We, therefore, propose this as a critical avenue for future research, enabling dynamic modeling of benefit–cost ratios under varying policy and market conditions to complement the current emissions and WTP findings.

3.3.6. Sensitivity Analysis

To enhance the robustness of our WTP estimates and address respondent heterogeneity, we conducted supplementary sensitivity analyses by recalculating mean WTP across key covariates from Table 5: household income, environmental knowledge (prior BD awareness), residence in the SMA, and age. The results, presented in Table 9, reveal systematic variations aligned with economic theory and prior non-market valuation studies.

Table 9. Sensitivity of mean willingness to pay (WTP) by covariates.

Variables [a]	Low Level [b]	Mean Level [b]	High Level [b]
	KRW 2 million	KRW 5.31 million	KRW 10 million
Income	KRW 3564.7	KRW 4688.8	KRW 6713.1
	(8.82)	(11.32)	(7.85)
	Don't know	8.3% know	Know
Knowledge	KRW 4368.1	KRW 4688.8	KRW 9454.1
	(10.98)	(11.32)	(6.79)
	Non-SMA [c]	53.3% SMA [c]	SMA [c]
Metro	KRW 3528.9	KRW 4688.8	KRW 5929.7
	(8.98)	(11.32)	(10.42)
	30-year-old	49.19-year-old	60-year-old
Age	KRW 5692.3	KRW 4688.8	KRW 4187.0
	(7.88)	(11.32)	(9.26)

Notes: [a] They are defined in Table 4. [b] Rows 1–3, as per variable show levels, mean WTP estimates, and *t*-values, respectively. [c] Indicates the Seoul Metropolitan Area. USD 1.0 was equivalent to KRW 1441.9 at the time of the survey.

Table 9 reports sensitivity analyses of mean WTP derived from the SB spike model coefficients (Table 5), disaggregating by household income, environmental knowledge (binary: know/don't know prior BD awareness), residence (SMA vs. non-SMA), and age cohorts. WTP escalates significantly with income—from KRW 3564.7 (USD 2.47 and *t*-value = 8.82) in the lowest tercile (KRW 2 million/month) to KRW 6713.1 (USD 4.65 and *t*-value = 7.85) in the highest (KRW 10 million/month).

Knowledgeable respondents display markedly higher WTP (KRW 9454.1, USD 6.55 and *t*-value = 6.79, +102% over mean level), highlighting information asymmetry as a barrier to policy support, while SMA residents—potentially exposed to higher urban pollution—express 26% elevated WTP (KRW 5929.7, USD 4.11 and *t*-value = 10.42). Conversely, older respondents (60 years) show reduced WTP (KRW 4187.0, USD 2.90 and *t*-value = 9.26, −10.7% vs. mean level), possibly reflecting risk aversion or lower future discounting horizons. These heterogeneities refine aggregate estimates (KRW 4688.8 base) and inform targeted interventions: subsidies calibrated to low-income/elderly groups could narrow the KRW 344.55 billion (USD 238.95 million) welfare gap described in Section 3.3.3, while awareness programs amplify total benefits to exceed 2030 costs under moderate diesel price scenarios. The sensitivity analysis (Table 9) effectively demonstrates WTP variability across household income, environmental awareness, residence (SMA), and age, underscoring heterogeneity in policy design implications.

Moreover, sensitivity analysis yields critical implications for diesel price fluctuations and subsidy scenarios, which is essential for robust policy design amid uncertain market conditions. Complementary scenario analyses simulate aggregate WTP under projected diesel price hikes of 10–20% by 2030 (from the current KRW 921 per liter baseline, as per Korea Petroleum Association forecasts), revealing that unadjusted household WTP (KRW 111.69 billion or USD 77.46 million annually) covers only 24.5% of the projected additional costs (KRW 456.24 billion or USD 316.41 million). This substantial shortfall—KRW 344.55 billion—underscores the need for targeted fiscal interventions, particularly given diesel's role in both transportation and industrial sectors, where price volatility could exacerbate regressive impacts on low-income and rural households with limited biofuel alternatives.

To bridge this gap, subsidy simulations demonstrate enhanced policy feasibility: reallocating fuel excise taxes (KRW 337.5 per liter) toward BD production incentives (e.g., KRW 200 per liter) could reduce BD premiums by 12.5% relative to PD (currently 81.3%), capping consumer price increases at under 5% and elevating WTP coverage to approximately 45%. Further integration of R&D measures—such as diversifying WCO imports or advancing marine microalgae feedstocks—could lower unit production costs from KRW 479.96 per liter to KRW 380 per liter, generating net social benefits that exceed outlays when incorporating CO_2 abatement values (KRW 146.8 billion annually). These strategies not only bolster public acceptance but align with international precedents, such as the EU Renewable Energy Directive III, affirming subsidies' cost-effectiveness in accelerating low-carbon fuel transitions while mitigating equity concerns.

3.4. Policy Pathways Derived from Public Concerns and WTP Findings

Respondent concerns from the 605 zero-WTP cases—53.1% institutional distrust ("taxes already paid"), alongside food security, raw material volatility, job losses, small and medium-sized enterprises (SMEs) protection, and supply chain risks—provide critical context for the aggregate WTP shortfall (KRW 111.69 billion vs. KRW 456.24 billion blending costs), revealing multidimensional barriers beyond price sensitivity [45–53]. These qualitative insights complement quantitative results, highlighting that while CO_2 benefits (1642 kt/year and KRW 146.8 billion social cost of CO_2 value) justify policy action, public acceptance hinges on addressing equity and supply stability, not merely environmental gains.

Actionable pathways emerge directly from this synthesis. First, prioritize domestic WCO/advanced feedstocks (microalgae via nuclear thermal discharge) to mitigate import risks noted by respondents, potentially creating 5000+ jobs while neutralizing food security concerns (minimal palm/soy impacts, as per Kristoufek et al. [45]). Second, reallocate diesel excise taxes (KRW 337/L potential) as tiered subsidies—low-income rebates and SME incentives—to bridge the KRW 344.55 billion gap, countering "tax fatigue" protests while preserving the SME ecosystem against refinery dominance. Third, launch targeted campaigns emphasizing co-benefits (air quality and energy security) validated by CV's spike model robustness.

This integrated approach—linking respondent insights to WTP/cost metrics—charts a pragmatic trajectory for the 8.0% mandate: pilot tiered incentives in 2026–2028, monitor via annual CV updates, and scale domestic production by 2030. Future extensions could model microalgae BD commercialization or cross-fuel blending policies, ensuring sustained public support amid nationally determined contribution decarbonization goals [48–53].

3.5. Limitations of the Study and Way Forward

Despite rigorous design, several limitations merit acknowledgment. First, while the survey achieved national representativeness through stratified random sampling across 17 administrative regions (KOSTAT 2020 Census strata, ±3% margin of error), interviewer-led face-to-face administration relied on a professional firm's 92%-validated callbacks rather than fully independent enumeration districts in every locality; perceptions of urban bias (53% SMA oversample reflecting diesel exposure) may persist, warranting future digital probability panels for geographic granularity. Second, the cross-sectional CV captures static 2025 preferences amid volatile diesel prices (±30% historical) and policy flux (post-2022 NDC revisions), potentially overstating long-term acceptance if salience fades or alternatives (e.g., electrification) gain traction—longitudinal replication is essential.

Third, hypothetical bias remains a CV caveat despite NOAA-compliant mitigations (budget reminders and pre-testing), with 94% protest zeros signaling institutional distrust that real referenda might amplify or attenuate via binding commitments. Fourth, the income tax vehicle proxies aggregate financing but abstract diesel-specific incidence (70–80% pass-through), understating support from non-users while conservative spike inclusion (vs. exclusion: +77% WTP) prioritizes credibility over optimism. Finally, unmodeled co-benefits (e.g., air quality such as nitrogen oxides or particulate matter reductions, quantified at KRW 50–100 billion via health valuations) likely elevate true societal benefits beyond CO_2-focused estimates, suggesting fuller integration in extensions.

The methodological framework employed in this study—particularly the OB CV approach integrated with the spike model to robustly handle zero-WTP responses—offers a transferable template for evaluating public acceptance of biofuel policy expansions beyond South Korea. Future research could fruitfully replicate this approach in other nations with comparable renewable energy mandates, such as EU member states under the Renewable Energy Directive (RED II/III), where advanced biofuel blending targets are legislated, or Latin American countries like Brazil and Argentina, which have longstanding ethanol blending programs. Specifically, applications could assess WTP for incremental blends of bioethanol (e.g., E10–E20 in gasoline), bioheavy oil (for marine or industrial decarbonization), or biojet fuel (aligned with sustainable aviation fuel initiatives under Carbon Offsetting and Reduction Scheme for International Aviation). By adapting bid structures and payment vehicles to local contexts—while retaining the spike model's capacity to model protest zeros and true-zero valuations—these extensions would generate policy-relevant benefit estimates, facilitate cross-country comparisons of public support, and inform cost–benefit analyses amid rising global biofuel ambitions. Such studies would also illuminate variations in WTP influenced by factors like feedstock availability, policy incentives, and energy security perceptions, ultimately advancing the evidence base for equitable energy transitions.

Moreover, future extensions of this research could incorporate longitudinal assessments to track the evolution of public acceptance of BD blending policies over time. Specifically, repeating CV surveys at 2–3-year intervals would enable monitoring of shifts in WTP as policy implementation progresses, technological advancements mature, and public awareness evolves. Complementing these economic valuations with thermogravimetric analysis (TGA) data and combustion kinetics modeling would provide technical validation of real-world emission reductions, bridging the gap between stated preferences and measurable environmental outcomes. For instance, panel data from repeated CV waves could be integrated with the empirical TGA profiles of B8 blends under varying operating conditions, allowing researchers to correlate changes in public WTP with validated reductions in CO_2, NOx, and particulate matter emissions. Such integrated longitudinal studies would offer policymakers dynamic evidence on the persistence of public support, informing adaptive strategies amid fluctuating fuel prices, regulatory adjustments, and competing low-carbon alternatives like electrification. This approach would strengthen the evidence base for sustained biofuel mandates, particularly in contexts where initial acceptance (as documented here) falls short of implementation costs.

Finally, integrating behavioral insights from prospect theory and nudge interventions could enrich future CV applications by addressing the 60.5% protest-zero prevalence observed here. Experimental designs incorporating gain-framed messaging (emphasizing co-benefits like energy security and air quality improvements) versus loss-framed appeals (highlighting climate damages avoided) would test framing effects on WTP distributions, while simplified payment ladders or social norm prompts could mitigate cognitive burdens in spike model responses. Complementing these with discrete CEs that decompose WTP into attributes—such as blending ratio increments, feedstock origin (domestic WCO vs. imported palm), and fiscal mechanisms (subsidies vs. taxes)—would enable marginal valuation of policy design elements, guiding regulators toward acceptance-maximizing blends. Such multidisciplinary extensions, validated against real-world policy pilots (e.g., phased B5→B8 rollouts), would ultimately translate empirical WTP into implementable decarbonization strategies, ensuring biofuel mandates align with both environmental imperatives and societal preferences.

4. Conclusions

In order to reduce CO_2 emissions from diesel vehicles, South Korea planned to increase the mandatory BD blending ratio for 2030 from the existing 5.0% to 8.0%. As a result of this increase, consumer prices for diesel will rise and there will need to be increased investment to produce more BD to meet the demand. For the increase in the mandatory BD blending ratio to be successful, it must be publicly acceptable. This study aimed to assess the public's acceptance by quantitatively analyzing the annual WTP per household for the increase using the data collected through a CV survey of 1000 households. The survey was commissioned from a professional polling company.

The SB spike model—designated as the main specification—applied to the 1000 observations secured robust statistical significance, with a Wald statistic p-value of 0.000, spike estimate of 0.606 (t-value = 39.29, closely matching the 60.5% empirical zero-WTP rate), and a mean household WTP of KRW 5052.3 (USD 3.50, t-value = 11.28, and 95% CI: KRW 4285.9–6072.4). Covariate extensions (Table 5) further confirmed intuitive influences—positive for income (t-value = 3.46), knowledge (t-value = 4.86), and urban residence (t-value = 4.86); negative for age (t-value = -1.98)—with pseudo-R^2 = 0.043 typical for cross-sectional CV data, validating the framework's reliability for policy valuation. These results underscore the methodological rigor underpinning the national WTP extrapolation (KRW 111.69 billion annually), providing policymakers with credible evidence amid blending costs of KRW 456.24 billion.

This study was significant not only from a policy perspective but also from a research perspective. From a policy perspective, the estimated annual average WTP was statistically significant. The value was KRW 5052.3 (USD 3.5) per year. At December 2024 price levels, the national version of the value reached KRW 111.69 billion (USD 77.46 million) per annum. The annual costs incurred for the increase in 2030 were KRW 456.24 billion (USD 316.41 million). The total annual WTP does not exceed the costs. Therefore, the authors judge that the increase is not yet sufficiently acceptable to the public. There is a need to improve public acceptance of the increase by lowering the price of BD. This can be achieved through implementing a government subsidy policy using the fuel taxes levied on diesel.

From a research perspective, CV, a well-established economic tool, was successfully employed for estimating the public's WTP for an increase in the mandatory BD blending ratio. To the best of our knowledge, this is among the first nationally representative studies to appear in the literature. Furthermore, the spike model was successfully utilized for effectively analyzing zero WTP response data. Statistically significant results were obtained. The framework adopted in this article could be transferred to analysis of similar issues in other countries. It could also be adopted to delve into public acceptance of increasing the blending ratio of biofuels other than BD, such as bioethanol, bioheavy oil, and biojet fuel.

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Abbreviations

The following abbreviations are used in this manuscript:

BAU	Business-as-usual
BD	Biodiesel
CE	Choice experiment
CI	Confidence interval
CO_2	Carbon dioxide
CV	Contingent valuation
DB	Double-bound
DC	Dichotomous choice
EU ETS	European Union Emission Trading System
GHG	Greenhouse gas
ICE	Internal combustion engine
KOSTAT	Korea Ministry of Data and Statistics
NOAA	National Oceanic and Atmospheric Administration
OB	One-and-one-half-bound
PD	Pure diesel
PP	Price premium
SB	Single-bound
SMA	Seoul Metropolitan Area
SME	Small and medium-sized enterprise
TB	Triple-bound
TGA	Thermogravimetric analysis
WCO	Waste cooking oil
WTP	Willingness to pay

Appendix A. One-and-One-Half-Bound Contingent Valuation Questions

According to analyses by relevant research institutions, the annual cost per household for increasing the mandatory biodiesel (BD) blending ratio ranges from KRW [Q1] to KRW [Q2].

Type A. Questions (Presented to half of respondents receiving [Q1, Q2])

A1. Is your household willing to pay an additional KRW [Q1] annually in income tax for the next 10 years to increase the mandatory BD blending ratio? If not paid, implementation will be difficult.

① Yes → [Go to A2]

② No → [Go to A3]

A2. Then, is your household willing to pay an additional KRW [Q2] annually in income tax for the next 10 years to increase the mandatory BD blending ratio? If not paid, implementation will be difficult.

① Yes → [Next page]

② No → [Next page]

Type B. Questions (Presented to the other half of respondents receiving [Q1, Q2])

A1. Is your household willing to pay an additional KRW [Q2] annually in income tax for the next 10 years to increase the mandatory BD blending ratio? If not paid, implementation will be difficult.

① Yes → [Next page]

② No → [Go to A2]

A2. Then, is your household willing to pay an even KRW [Q1] annually in income tax for the next 10 years to increase the mandatory BD blending ratio? If not paid, implementation will be difficult.

① Yes → [Next page]

② No → [Go to A3]

A3. Then, does your household have no willingness to pay even KRW 1?

① Willing to pay a small amount → [Next page]

② No willingness to pay at all → [Go to A4]

A4. What is the most important reason your household refuses additional income tax for the BD blending ratio increase? (Single selection)

① Expansion of BD supply should be carried out with taxes already paid

② There is not enough information provided in the questionnaire to decide whether to pay

③ BD is not important enough to be given priority

④ The additional taxes I pay will not be used to expand the supply of BD

⑤ BD is not of interest to my household

⑥ The government is already spending a lot of money on BD

⑦ Expanding the supply of BD is of little value to me

⑧ Our household cannot afford to pay for the expansion of the BD supply

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